409 [*Black Holes: The Membrane Paradigm*.] Thorne, Price, and Macdonald (1986). See also Price and Thorne (1988).

CHAPTER 12

412 General comment about Chapter 12: The historical aspects of this chapter are based on (i) my own personal experience as a participant, (ii) my interviews with other participants (INT-DeWitt, INT-Eardley, INT-Hartle, INT-Hawking, INT-Israel, INT-Penrose, INT-Unruh, INT-Wald, INT-Wheeler, INT-Zel'dovich), (iii) my reading of scientific papers the participants wrote, and (iv) the following published accounts of the history: Bekenstein (1980), Hawking (1988), Israel (1987).

412 [The Idea hit . . . so quickly.] This and the subsequent description of how Hawking arrived at the idea come from INT-Hawking and Hawking (1988). Hawking published the details and consequences of his idea, as sketched in the first section of this chapter, "Black Holes Grow," in Hawking (1971b, 1972, 1973).

414 [following Roger Penrose's lead,] Penrose (1965).

414 Box 12.1: Hawking (1972, 1973).

417 [Stephen Hawking was not the first . . . Werner Israel] INT-Israel, INT-Penrose, INT-Hawking.

417 [Penrose's amazing 1964 discovery . . . singularity at its center.] Penrose (1965).

418 Box 12.2: Hawking (1972, 1973).

419 [Hawking and James Hartle . . . gravity of other bodies.] Hawking and Rartle (1972).

422 [Demetrios Christodoulou . . . equations of thermodynamics.] Christodoulou (1970).

425 [Jacob Bekenstein was not persuaded.] Bekenstein describes this and the controversy with Hawking that followed in Bekenstein (1980). Bekenstein published his black-hole entropy conjecture and his arguments for it in Bekenstein (1972, 1973).

426 [Les Houches summer school,] The proceedings of the 1972 summer school were published in DeWitt and DeWitt (1973).

427 [Bardeen, Carter, and Hawking . . . *laws of black-hole mechanics*] Bardeen, Carter, and Hawking (1973).

428 [Zel'dovich had brought me to Moscow . . .] Charles Misner and John Wheeler accompanied me on my June 1971 visit to Moscow, but they were not with me at Zel'dovich's apartment during the discussion described in the following paragraphs.

429 [Zel'dovich, his eyes dancing, . . .] I have reconstructed the following conversation from memory, and have translated it into less technical language than we actually used.

433 [Zel'dovich, however, did not forget; . . . his paper was published] Zel'dovich (1971).

434–435 [Starobinsky described Zel'dovich's conjecture . . . does, indeed, radiate.] Zel'dovich and Starobinsky (1971).

435 [Then came a bombshell.] Hawking describes, in Hawking (1988), how he arrived at his "bombshell" discovery that all black holes radiate. He published the discovery and its implications in Hawking (1974, 1975, 1976).

437 [This and the demand for a perfect mesh,. . . almost completely.] See, e.g., Wald (1977).

437 Footnote 11: Wald (1977).

439 [Perhaps the simplest. . . particles rather than waves:] Hawking (1988).

442 [Gradually . . . new understanding embodied in Figure 12.3.] Chapter 8 of Thorne, Price, and Macdonald (1986), and references therein.